

Considering the latest UN report on Water Bankruptcy Era, can't AI data centers just consume sea water or any other water that isn't drinkable?

Posted by u/roguecolor • 0 points • 16 comments

Why is this such a problem? Genuine question

Comments

u/glitteraffe • 1 points • 2026-01-30 12:38 UTC

YES !!!!

Data centers worsen the strain via traditional evaporative cooling's heavy freshwater use, yet solutions exist! Advanced liquid cooling (direct to chip, immersion, closed loop systems) recirculates fluid with minimal to zero evaporation, achieving near-zero ongoing consumption while powering dense AI workloads.

Governments still offer subsidies, tax breaks, and lax rules that fuel hyperscale growth without hydrological limits, favoring digital expansion over resource security.

This shortsighted policy drives geopolitical risks: intensified resource competition in stressed areas sparks unrest, strains transboundary pacts, and turns water into leverage in trade disputes or migration crises, threatening food, energy, and stability.

u/Superior_Mirage • 2 points • 2026-01-28 06:44 UTC

The main thing is that, out of all the problems AI might cause, water isn't one of them.

It uses somewhere around 1-2% of the water turf (lawns, golf courses, etc.) does in the U.S. -- if we need water, that's where to stop wasting. Even the worst-case projections never have AI overtaking that.

That doesn't mean that it can't cause water problems locally, if municipalities greenlight data centers without actually making sure their utilities can handle it. But with everything else going on in the world, making sure idiots don't get elected locally is relatively easy.

u/Temporary_Double8059 • 5 points • 2026-01-27 17:34 UTC

Water is utilized for heat dissipation and therefore evaporated leaving behind minerals. Salt Water in particular is corrosive to start with and will leave lots of salt deposits as its ran through the system.

u/Ghigs • 6 points • 2026-01-27 17:15 UTC

To use sea water you need to filter it and then have all stainless steel pipes. It can be done, some nuclear power plants use seawater. But it's expensive.

Also the whole "AI - water" trope is overdone. It's like nothing compared to agriculture use.

u/roguecolor • 2 points • 2026-01-27 17:17 UTC

Yeah I know, there are many other issues regarding water consumption at the moment, my main doubt at the time was really the type of water that can be used by AI data centers aside from clean water

u/Spiritual_Panic_6992 • 1 points • 2026-01-28 10:03 UTC

Mainly used for cooling. The loss comes from evaporation, which is why seawater cannot be used. After the water evaporates, impurities in the seawater will precipitate and condense inside the pipeline, causing the entire pipeline to be scrapped within a few years.